

Database Management System

CSIT — 4th Semester — 2082 — Answer Sheet
Subject Code: CSC265

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Consider an online Bookstore, from which customers can buy book. Each customer can place multiple orders. Each order has one or more books. The Bookstore needs to store information about each customers, their orders, and the books available in the store.a) Draw an ER diagram for the online Bookstore system including appropriate attributes, keys, and relationships.b) Convert the ER diagram into a relational schema. Show all the primary keys and foreign keys.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Consider the following relational schema:Student(SID, SName, Dept, Year)Course(CID, CName, Credit)Enroll(SID, CID, Grade)Specify the following queries in both SQL and Relational Algebra on this database schema:a) List names of students from the "CSIT" department whose name starts from A.b) Display student names and course names they have enrolled in.c) Count total students enrolled in each course.d) Update grade 'B' to 'A' for all students in "DBMS".e) Delete records of students whose Year is 2024.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define Normalization. Explain 1NF, 2NF and 3NF with a suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Differentiate between traditional file processing system and database approach.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain the concepts of schema, instance, and data independence.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define relational constraints. Explain domain, entity integrity, and referential integrity with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain the basic operations of relational algebra: UNION, INTERSECTION, DIFFERENCE, and CARTESIAN PRODUCT with suitable examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Explain different types of functional dependencies with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain ACID properties in database.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is concurrency control? Explain optimistic and multiversion concurrency control techniques.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is a schedule in transaction processing? Explain how concurrency control ensures serializability.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Explain any one database recovery technique.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.